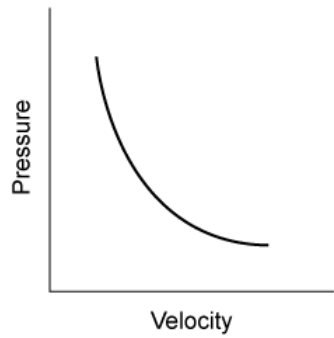


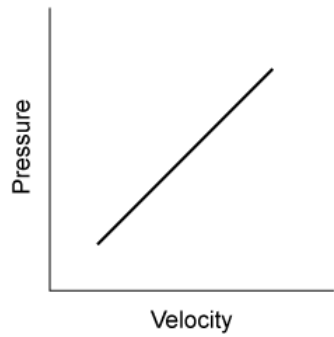
An ideal fluid flows within a fixed tube with decreasing cross-sectional area. Assuming constant height and volumetric flow rate, which of the following graphs best illustrates the relationship between fluid pressure and the increase in fluid velocity?

✔ ☒ A.



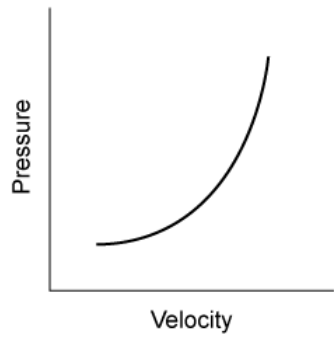
[50%]

☐ B.



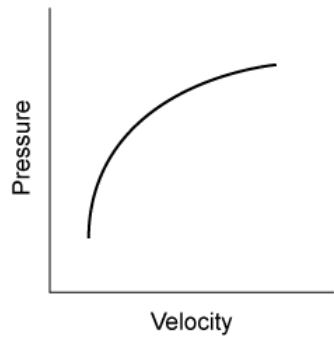
[21%]

☐ C.



[20%]

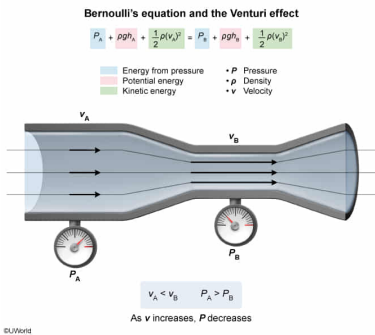
☐ D.



[8%]

Correct
50% answered correctly

Explanation:



One application of **Bernoulli's equation** relates to the travel of fluids between two points (A and B) within a fixed conduit (a pipe or channel through which a fluid passes). Each side of Bernoulli's equation contains three terms, the first of which is pressure (P), a measure of [energy per unit volume](#). The second and third terms reflect the potential and kinetic energy of a fluid, which both differ from the potential and kinetic energy of solids by being related to fluid **density (ρ)** rather than to mass:

$$P_A + \rho gh_A + \frac{1}{2} \rho v_A^2 = P_B + \rho gh_B + \frac{1}{2} \rho v_B^2$$

Bernoulli's equation is useful in modeling the flow of [ideal fluids](#) because ideal fluids are non-compressible (ie, unchanging density). Fluid incompressibility requires that changes in conduit geometry (ie, cross-sectional area) cause changes in fluid velocity because the [volumetric flow rate](#) through interconnected segments of a conduit must be equal. Consequently, changes in [conduit geometry](#) are accompanied by changes in fluid pressure or potential energy, maintaining the conservation of energy.

In this question, an ideal fluid flows through a fixed tube with a decreasing cross-sectional area. Because the height (h) remains constant, the potential energy terms can be eliminated from Bernoulli's equation:

$$P_A + \frac{1}{2} \rho v_A^2 = P_B + \frac{1}{2} \rho v_B^2$$

Further rearrangement of the remaining terms yields an expression that relates the difference in pressure at both points to the difference in kinetic energy at both points:

$$P_A - P_B = \frac{1}{2} \rho (v_B^2 - v_A^2)$$

Because the cross-sectional area of the conduit decreases while the volumetric flow rate remains constant, fluid velocity must increase:

$$\frac{1}{2} \rho (v_B^2 - v_A^2) > 0$$

Substituting this inequality into the previous arrangement of Bernoulli's equation yields an expression for pressure:

$$P_A - P_B > 0$$

$$P_A > P_B$$

Therefore, the pressure of ideal fluid in motion will decrease as fluid velocity increases (**Choice A**), a phenomenon known as the **Venturi effect**. The Venturi effect has relevance to practical scenarios (eg, aviation, blood flow) and can be observed using a device known as a [Venturi tube](#).

Educational objective:

The Venturi effect describes the decrease in pressure observed concurrent to an increase in fluid velocity. The Venturi effect is modeled by Bernoulli's equation, which describes the conservation of energy in fluids.

Which of the following changes will increase the rate at which a mammalian organism loses body heat to the environment?

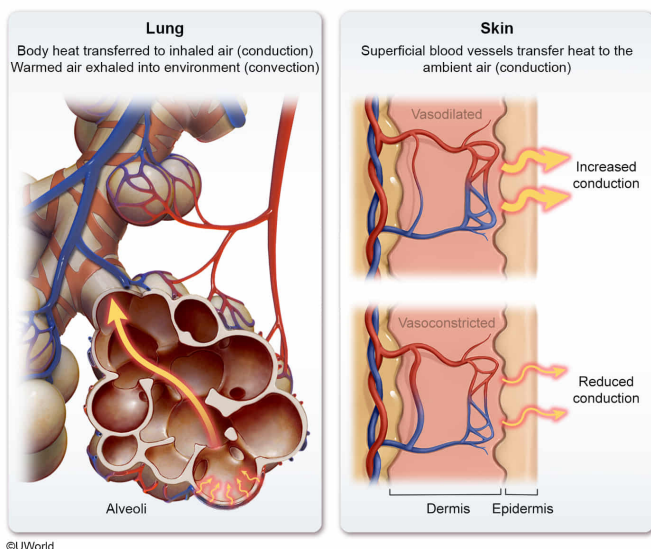
- ☐ A. Vasoconstriction of the organism's superficial blood vessels [12%]
- ☐ B. Vasoconstriction of the organism's pulmonary blood vessels [12%]
- ☐ C. Replacement of the organism's lean muscle mass with superficial fat [4%]
- ✔ ☒ D. Increase in the organism's rate of pulmonary ventilation [70%]

Correct

71% answered correctly

Explanation:

Biological mechanisms of environmental heat transfer



Heat transfer is the transmission of thermal energy between two objects or systems by processes such as conduction and convection. In **conduction**, thermal energy passes between two objects (or domains) through physical contact. During convection, the movement of a fluid, such as blood or air, serves to transport heat from warmer regions to cooler regions. Heat transfer between two objects or regions occurs until the temperature equalizes.

The physical dimensions of the transfer interface influence the rate of heat transfer (H) between objects. For example, the rate of heat transfer between two objects connected by a **thermal conductor** depends on the conductor length (L), conductor area (A), and difference between the temperature of the hotter object (T_H) and the temperature of the cooler object (T_C):

$$H \propto \frac{A(T_H - T_C)}{L}$$

In biological contexts, heat transfer to the environment is facilitated by processes like ventilation (ie, inhalation and exhalation) and the flow of blood through superficial anatomical structures. The regulation of superficial blood

flow via **vasoconstriction and vasodilation** adjusts blood flow to the skin, which is in physical contact with the ambient air. Therefore, changes in skin blood flow modulates conductive heat loss to the environment.

Heat transfer to the environment through **ventilation** occurs through a similar mechanism except that conductive heat transfer to inhaled air is followed by convective heat transfer through exhalation. Therefore, increasing the rate of respiration increases the rate of heat loss to the environment.

(Choice A) Vasoconstriction would reduce blood flow to superficial areas and *diminish* the rate of heat loss to the environment.

(Choice B) Vasoconstriction would reduce blood flow to the lungs and *diminish* the rate of heat loss due to respiration.

(Choice C) Replacing lean muscle mass with superficial fat would increase the thickness of the superficial structures that act as thermal insulators, decreasing the rate of heat loss to the environment.

Educational objective:

Heat transfer from an organism to the environment can be accomplished by increasing superficial blood flow (enhancing thermal conduction) or by increasing respiration (enhancing conduction and convection).

Time spent:0 sec

Subject:
Physics

QID:402365

System:
Fluids